Project owner support :

C.E.S. International

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Project owner :

MKR CORP
WHERE MOBILITY MEETS RESPONSIBILITY

Campinas, Sao Paulo in Brazil

VBP - ElectroPark Brazil

PERSPECTIVE

BIM execution plan for the ElectroPark Brazil project

Index E

Phase	Index	Date	Comments	Author	Approved by
Preliminary	**A**	November 14th 2025	Front cover	DE OLIVEIRA	Team 4
Preliminary	**B**	November 17th 2025	Presentation, goals and uses	DE OLIVEIRA	Team 4
Preliminary	**C**	November 19th 2025	milestones and organisation	DE OLIVEIRA	Team 4
Preliminary	**D**	November 21th 2025	BIM PROCESSES & Information management	DE OLIVEIRA	Team 4
Preliminary	**E**	November 26th 2025	Information management and CDE	DE OLIVEIRA	Team 4

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CHAPTER 1 GENERAL PRESENTATION

1.1 Purpose of this document

The BIM Execution Plan (BEP) serves to clarify BIM objectives, define collaborative processes, and structure information exchanges so that all stakeholders operate within a shared and consistent framework. It sets out the approach proposed by the prospective lead appointed party (CES International) for the execution, monitoring, and control of BIM throughout the project, in response to the appointing party's (MKR) requirements.
In addition to international BIM standards and ISO 19650 principles, all BIM processes, deliverables, and data structures described in this BEP must also comply with the applicable ABNT (Associação Brasileira de Normas Técnicas) standards, ensuring full alignment with Brazilian regulatory requirements.

To ensure coherent, coordinated, and high-quality BIM production, the BEP outlines the following aspects:

BIM Management
- Presentation of the BIM team
- Roles and responsibilities of contributors

- BIM Objectives
  - Definition of BIM objectives
  - Level of development (LOD) of models per phase based on BIM uses
  - Model review procedures

- Collaboration
  - Collaboration strategy
  - Information exchange platform
  - Revision frequency

- Publication
  - Deliverable management
  - Procedures for submission, review, and approval

1.2 Building Information Modeling (BIM)

The term BIM, also known as Building Information Modeling, refers to a structured and organized database containing all information related to a project. BIM enables all project stakeholders to exchange and share data in a transparent and efficient manner. It is based on one or several multidimensional models—3D, 4D (time), and 5D (cost)—in which extensive information can be assigned to each object in the model through a set of attributes.
A digital model evolves over time, not only in scale and complexity but also in the scope of its applications and uses throughout the entire lifecycle of a project, from design to maintenance. There is no single or universal BIM; rather, BIM is a working methodology that must be adapted to the expectations and requirements of the client.

1.3 Reference Documents

The preparation of this document is made possible through the following reference materials:
- Guide méthodologique pour des conventions de projets en BIM - Version 2 Mars 2018
- MKR corp. BIM requirements
- Memorandum of Understanding (MoU)
- Term of Reference (ToR)
- A BIM convention for CESI Nanterre as a template

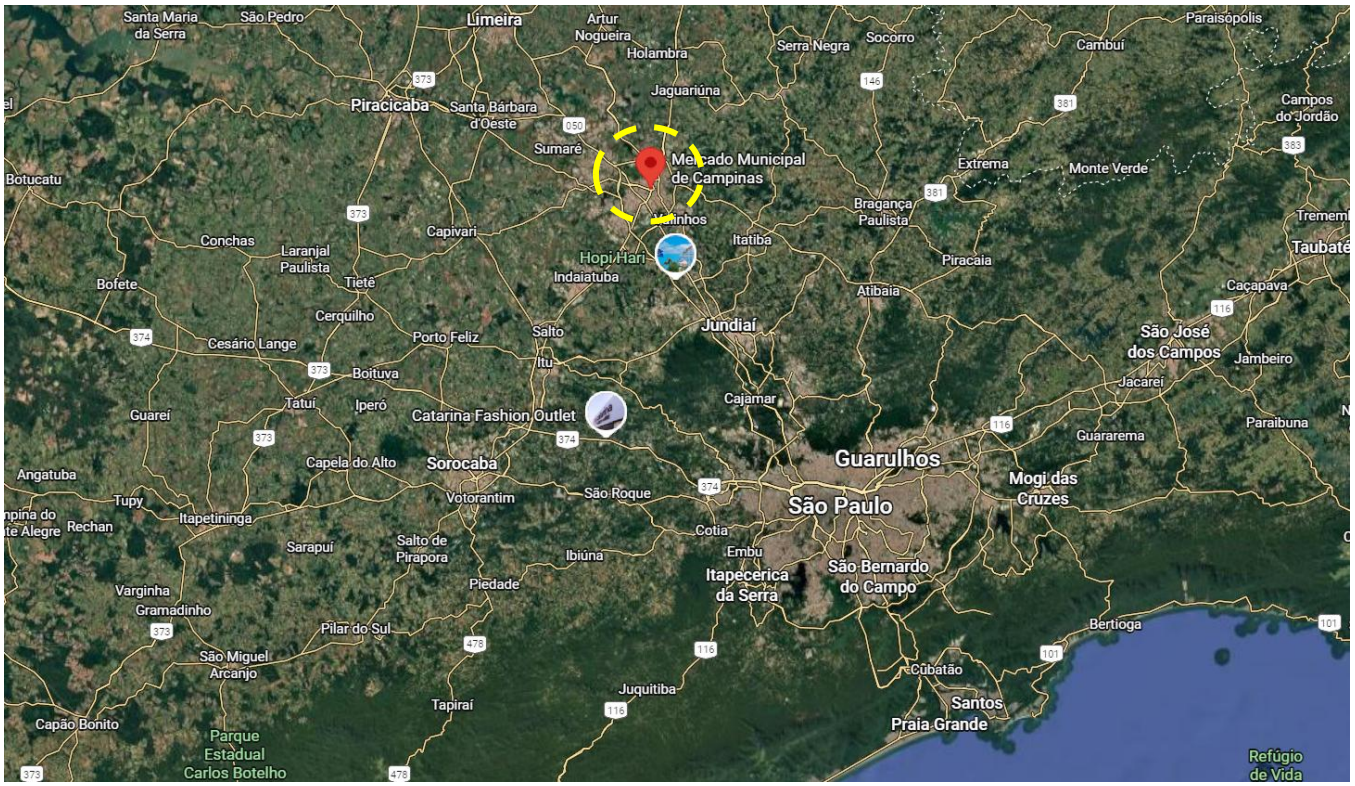
Chapter 2 PRESENTATION AND GOALS

2.1 Project name:

Very Big Project (VBP), ElectroPark Brazil

2.2 Project location:

Campinas, São Paulo in Brazil



2.2 Project code:

The selected project code is : VBP

2.3 Project presentation

ElectroPark Brazil is a multifunctional logistics and vehicle processing hub under development in Campinas, São Paulo. Serving as a post-production interface between MKR's vehicle assembly lines and national and international distribution partners, the facility is designed as a strategic buffer zone. Covering approximately 20,000 m², it will accommodate up to 1,200 electric vehicles while providing integrated infrastructure for charging, inspection, quality control, and digital supervision.

The site will feature a multi-level parking structure with photovoltaic integration, equipped with both rapid and slow AC/DC charging stations, secure access control, and digital tracking systems for all inbound and outbound vehicle flows. Construction is scheduled for completion by December 2026, with commissioning targeted for March 2027. BIM will be applied at Level 2 maturity to ensure coordinated modeling, structured data sharing, and interdisciplinary collaboration throughout the project lifecycle.

The project proposal must consider the following key parameters to ensure design and operational feasibility:

- Site area: 22,000–25,000 m²
- Vehicle capacity: minimum 900 units
- Maximum structure height: 15 m (local zoning restriction)
- Soil bearing capacity: to be confirmed; assume moderate strength

- Drainage and flood protection: critical due to the high-rainfall tropical climate
- Standards and compliance: adherence to ABNT, international safety codes, and accessibility regulations

2.4. Project schedule and milestones

Milestones	Deadline
Preliminary studies	April 11th 2025
Schematic design	November 28th 2025
Detailed design	November 28th 2025
Construction documentation	January 26th 2026
Construction	December 12th 2026
Handover	April 15th 2027

Table 1 – Project milestones

CHAPTER 3 BIM Conception TEAM

Any party involved in the project is regarded as part of the BIM team once it can provide information essential to fulfilling one of the use cases outlined in this convention.

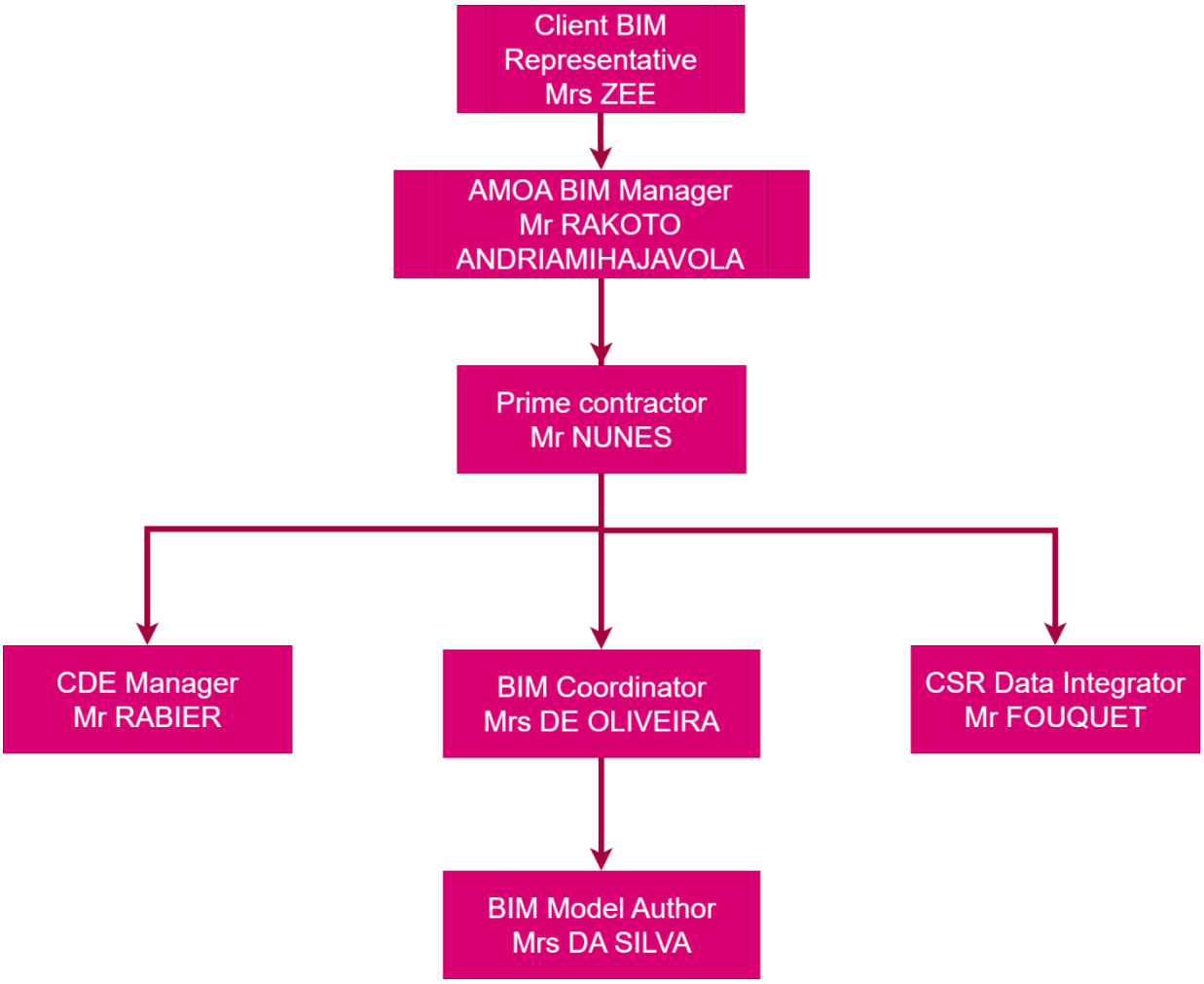
3.1 Roles and actors list:

Name	role	company		mail	phone number
		name	code		
Linlin ZEE	Client BIM Representative	MKR Corp	MOA	L.zee@mkr.com	0612345670
Maixent RAKOTO ANDRIAMIHAJAVOLA	AMOA[1] BIM Manager	CES International	AMOA	R.maixent@cesi.com	0612345672
Maxime FOUQUET	CSR[2] Data integrator	CES International	AMOA	M.fouquet@cesi.com	0612345674
Gabriel RABIER	CDE Manager	CES International	AMOA	G.rabier@cesi.com	0612345675
Lorena DE OLIVEIRA	BIM coordinator	BEC construction	BEC	L.deoliveira@becconstructionlr.com	0612345676
Miguel DA SILVA	BIM modeler	BEC construction	BEC	M.dasilva@becconstructionlr.com	0612345676
Augusto NUNES	Prime contractor	BEC construction	BEC	A.nunes@becconstructionlr.com	0612345676

[1] Owner's reprentative

[2] Corporate Social Responsibility (CSR) refers to the way a company integrates ethical, social, and environmental concerns into its business operations and interactions with stakeholders.

3.2 BIM stakeholders OBS:



3.3 Roles and Responsibilities

The BIM processes implemented for this project are structured around three levels of responsibility, in accordance with good practices aligned with ABNT and the Brazilian adoption of the ISO 19650 series (ABNT NBR ISO 19650):

- BIM Manager, supporting project leadership
- BIM Coordinator, assigned to each discipline
- BIM Modeller, responsible for model and deliverable production

There are also three different stakeholders outside of the core BIM process but essential for the project: CDE Manager, CSR Integrator, and the Prime Contractor.

3.3.1 AMOA BIM Manager

The BIM Manager serves as the project lead for all matters related to digital models, information management, and BIM methodology. This role ensures that BIM processes comply with the requirements defined in ABNT[3] NBR[4] ISO 19650 and with project-specific standards.

Main duties:

- Establish and maintain BIM processes, including:

[3] The Brazilian Technical Standards Association (Portuguese: *Associação Brasileira de Normas Técnicas*, ABNT)
[4] *Norma Brasileira* and abbreviated NBR

  - Collaboration workflows between all project stakeholders
  - Levels of Information Need (equivalent to LOD/LOI) required per phase
  - BIM templates, naming conventions, and modelling standards
  - Codification of BIM files and structured information

- Oversee the coordination and integration of data across disciplines
- Ensure compliance with the approved procedures, protocol, and standards
- Guarantee model interoperability across the software platforms used on the project
- Maintain technological monitoring of relevant BIM tools and methods
- Combine and federate discipline-specific models into a coordinated project model
- Supervise the discipline BIM Coordinators

Responsibilities:

- Custodian[5] of BIM procedures and standards
- Custodian of collaboration processes and data exchanges
- Custodian of the correct implementation of the BIM Execution Plan

3.3.2 BIM COORDINATOR

The BIM Coordinator acts as the main liaison between the BIM Manager and their respective discipline team, ensuring that their discipline complies with ABNT-aligned BIM processes.

Main duties:

- Apply and enforce the project's BIM procedures within their discipline
- Lead and monitor BIM production activities
- Support and guide the BIM Modellers
- Develop discipline-specific BIM content
- Verify compliance with the BIM Execution Plan
- Publish and distribute discipline models within the Common Data Environment (CDE)

Responsibilities:
Custodian of the correct application of BIM procedures for their discipline

3.3.3. BIM MODELER

The BIM Modeller produces the discipline models and associated digital deliverables in accordance with the project's BIM standards and the requirements set out in ABNT NBR ISO 19650.

Responsibilities:

- Report modelling progress and issues to the BIM Coordinator
- Produce discipline models in full compliance with the BIM Execution Plan, the BIM protocol, and all associated annexes
- Ensure modelling accuracy, consistency, and alignment with the established standards

NB: BIM modelers and BIM coordinator of the project are the same person because the models are less complicated than usual.

3.3.4 CDE Manager

The CDE Manager is responsible for managing the Common Data Environment (CDE), ensuring that all project information is stored, organized, and shared according to ABNT ISO 19650 principles.

[5] Keeper, guardian of

Main duties:

- Set up, maintain, and monitor the CDE platform
- Control access permissions for all stakeholders
- Ensure that model and document versions are properly tracked
- Support BIM teams with uploading, downloading, and sharing information
- Ensure compliance with CDE protocols and project information standards

Responsibilities:

- Custodian of the project's CDE
- Ensures secure, structured, and controlled data exchange among all parties

3.3.5 CSR Integrator

The CSR Integrator is responsible for coordinating the project's Corporate Social Responsibility (CSR) requirements and sustainability reporting, ensuring that BIM and project data support environmental, social, and governance objectives.

Main duties:

- Collect and integrate sustainability data from BIM models (materials, energy, LCA, carbon footprint)
- Verify compliance with project CSR goals and standards
- Provide structured reports for stakeholders on sustainability indicators
- Liaise with the BIM team to ensure data is correctly captured in models

Responsibilities:

- Custodian of CSR-related data within the BIM environment
- Ensures sustainability targets and KPIs are traceable and verifiable

3.3.6 Prime Contractor

The Prime Contractor is responsible for overall project delivery, coordinating subcontractors, and ensuring compliance with contractual, technical, and safety requirements.

Main duties:

- Oversee construction execution and schedule management
- Ensure project quality, safety, and compliance with design specifications
- Interface with the BIM Manager & Coordinators, CDE, and CSR Integrator to receive coordinated information for execution

Responsibilities:

- Project delivery and coordination
- Ensures integration of BIM, CDE, and CSR outputs into site operations

3.4 RACI Matrix

Task / Uses — Stakeholders	Client BIM Representative	AMOA BIM Manager	BIM Coordinators	BIM Modeler	Prime contractor	CDE Manager	CSR Data Integrator
BIM Execution Plan	A	A	R	I	I	C	C
3D Design & Coordination	A	A	R	R	R	A	C
Clash Detection	I	R	I	I	A	A	I
Model-Based Documentation	I	I	R	R	R	A	I
4D / Site Logistics Simulation	C	A	I	I	R	A	A
Asset Data Structuring	A	A	R	R	R	A	A
Sustainability Analysis	I	C	C	C	A	A	A
5D Cost Simulation	C	C	R	R	R	A	I
6D Facility Management Integration	I	C	R	R	R	A	A
7D Lifecycle & Decommissioning	I	C	R	R	R	A	C

RACI Legend:

- R – Responsible
  Executes the task and produces the deliverables.
- A – Accountable
  Ultimately answerable for the task; validates and approves the final outcome.
- C – Consulted
  Provides input, expertise, or feedback during the task.
- I – Informed
  Kept updated on progress or decisions but does not participate in the task.

CHAPTER 4 BIM GOALS AND USES

4.1 BIM goals

These are the general BIM goals of the project:

- Improve coordination between disciplines (architecture, structure, MEP, IT, energy systems)
- Reduce errors and conflicts through early digital clash detection
- Increase transparency and traceability of project data across phases
- Facilitate planning, budgeting, and phasing through better visualization and simulation
- Prepare structured information for the operational phase and asset management

These objectives are to be monitored through regular coordination sessions, quality reviews, and the use of agreed Key Performance Indicators (KPIs) as defined in the BEP.

These are the technical BIM goals of the project:

- Delivery of federated models, updated at key milestones
- Generation of documentation (plans, schedules, quantities) directly from models
- Maintenance of model integrity through version control, naming conventions, and
- IFC compliance
- Facilitation of interdisciplinary reviews through compatible file formats (native +
- IFC 4.3)
- Provision of LOD-compliant geometry and data attributes per project stage
- Compatibility with downstream tools (e.g., CAFM, CMMS, energy dashboards)

Where possible, modelling choices should support performance-based design strategies and predictive analysis.

4.2 BIM targeted uses

No.	BIM Use	Description
1	3D Design and Coordination	The process consists of creating discipline-specific models (architecture, structure, MEP, energy systems, IT) and ensuring their compatibility for use in federated model[6] reviews. The intent is to support coordinated design development and reduce inconsistencies between trades.
2	Clash Detection	The process consists of identifying spatial conflicts and inconsistencies by federating models from all disciplines. Regular coordination sessions are held to detect, track, and resolve clashes using model federation tools.
3	Model-Based Documentation	The process consists of producing drawings, diagrams, schedules, and quantity take-offs directly from the discipline models. This ensures consistency between documentation and the models and automates updates triggered by design changes.
4	4D / Site Logistics Simulation	The process consists of visually sequencing construction activities, site organization, and phasing using time-based simulation tools. This BIM use is optional but recommended to support construction planning and logistics optimization.
5	Asset Data Structuring	The process consists of integrating object-level information (attributes, classification, maintenance data) into the model to support future building operation, asset management, and data handover.

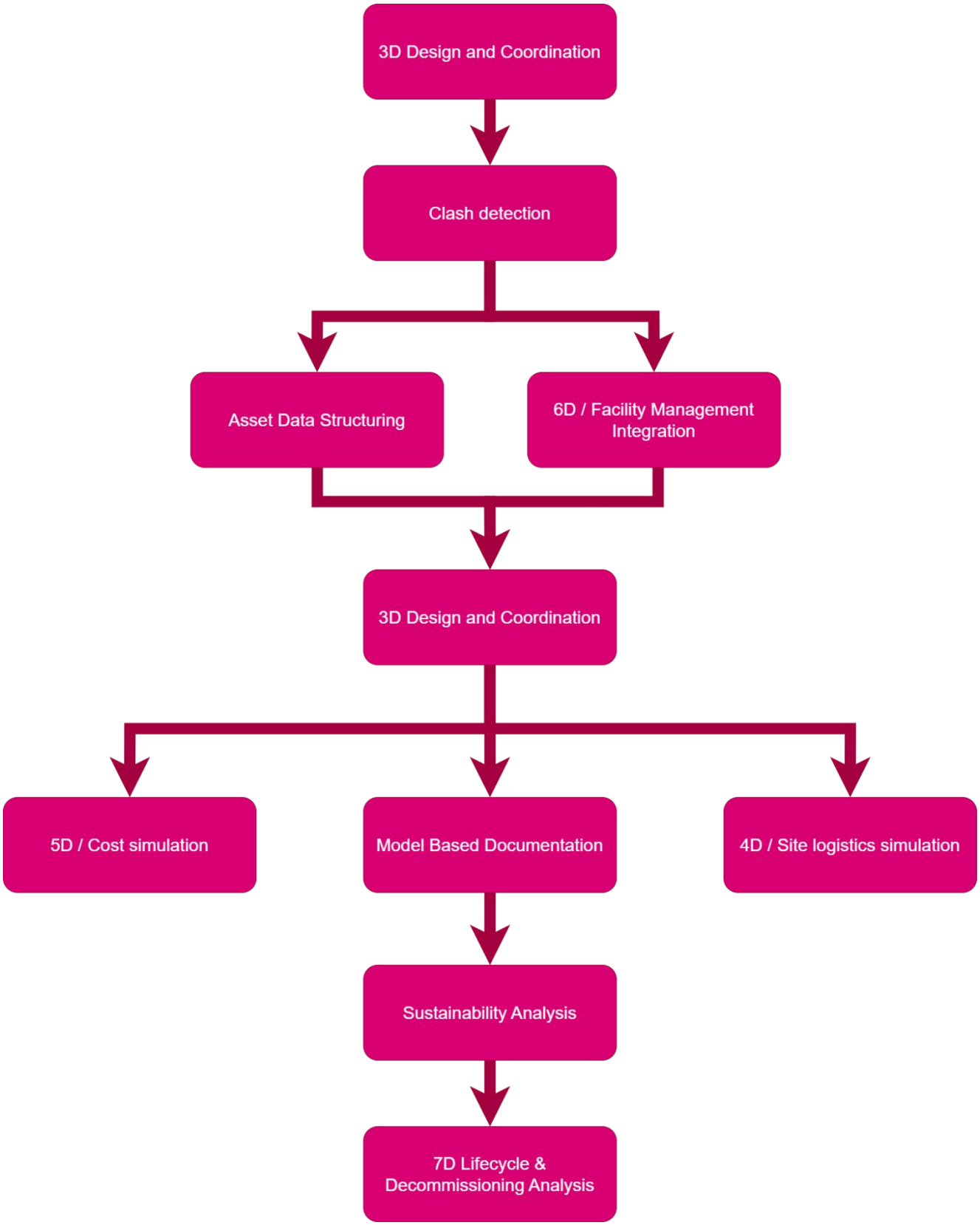
[6] Federated models are separate BIM discipline models (architecture, structure, MEP, etc.) combined into a single, coordinated model. This allows all teams to see conflicts, overlaps, and interactions between systems without changing each discipline's original model. For example: the software *NAVISWORK* does this

6	Sustainability Analysis	The process consists of linking the model to simulation tools for embodied carbon assessment, energy modeling, environmental impact evaluation, and CSR performance tracking.
7	5D Cost Simulation	The process consists of integrating quantity take-offs and cost data with the model to simulate budgets, track cost evolution, and support financial decision-making. The model is linked to unit prices and updated automatically when design changes occur.
8	6D Facility Management Integration	The process consists of enriching the model with asset information required for building operation. This includes equipment data, maintenance schedules, warranties, and classification systems to ensure a structured and interoperable digital handover compatible with facility management tools.
9	7D Lifecycle & Decommissioning Analysis	The process consists of using the model to support long-term lifecycle strategies, including end-of-life planning, recyclability assessment, material recovery, and deconstruction sequencing. This BIM use enables full lifecycle evaluation and supports circular economy objectives.

BIM Objective	Associated BIM Use(s)	BIM Use Name(s)
Improve coordination between disciplines (architecture, structure, MEP, IT, energy systems)	01, 02	3D Design and Coordination; Clash Detection
Reduce errors and conflicts through early digital clash detection	02	Clash Detection
Increase transparency and traceability of project data across phases	03, 05	Model-Based Documentation; Asset Data Structuring
Facilitate planning, budgeting, and phasing through better visualization and simulation	04, 07, 09	4D / Site Logistics Simulation; Extended Simulation (5D–7D)
Prepare structured information for the operational phase and asset management	05, 06, 08, 09	Asset Data Structuring; Sustainability Analysis; Extended Simulation (6D–7D)

CHAPTER 5 BIM PROCESS

5.1 General process



5.2 BIM use's processes

5.2.1 3D Design and coordination

Description:

The process consists of creating discipline-specific models (architecture, structure, MEP, energy systems, IT) and ensuring their compatibility for use in federated model[7] reviews. The intent is to support coordinated design development and reduce inconsistencies between trades.

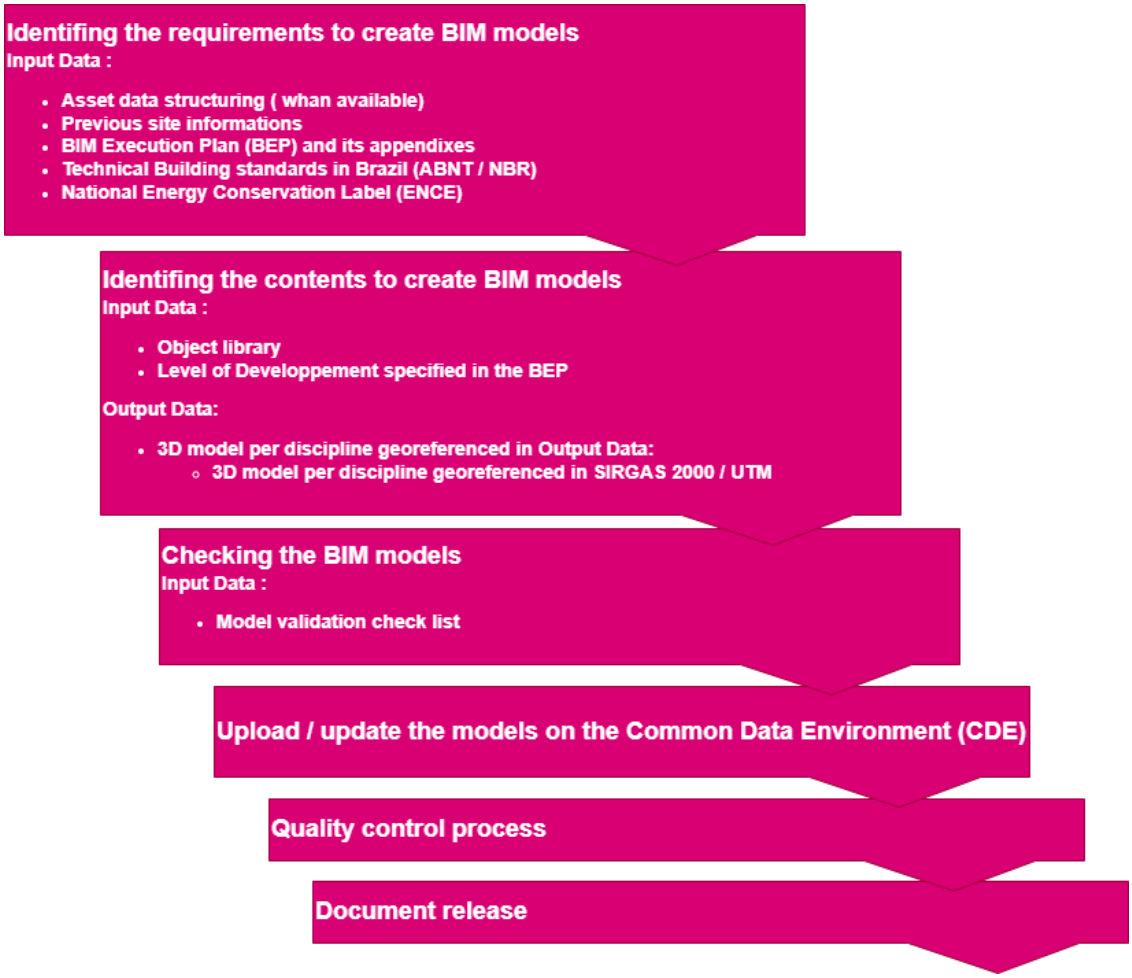
Benefits:

- Having a common model for every stakeholder of the project
- Better control on the project's scope (cost, delay and conception)
- Better visualization of the final outcome
- Genuine collaboration between stakeholders
- Improving the conception's reliability

Skill requirements:

- Knowledge and experience in construction
- Ability and mastery to use, modify and complete a 3D BIM model

Procedure:



[7] Federated models are separate BIM discipline models (architecture, structure, MEP, etc.) combined into a single, coordinated model. This allows all teams to see conflicts, overlaps, and interactions between systems without changing each discipline's original model. For example: the software *NAVISWORK* does this

5.2.2 Clash Detection

Description:

The process consists of using clash-detection software during model coordination to identify conflicts by comparing the 3D models of the different construction systems. The objective is to detect geometric interferences early and support efficient multidisciplinary coordination.

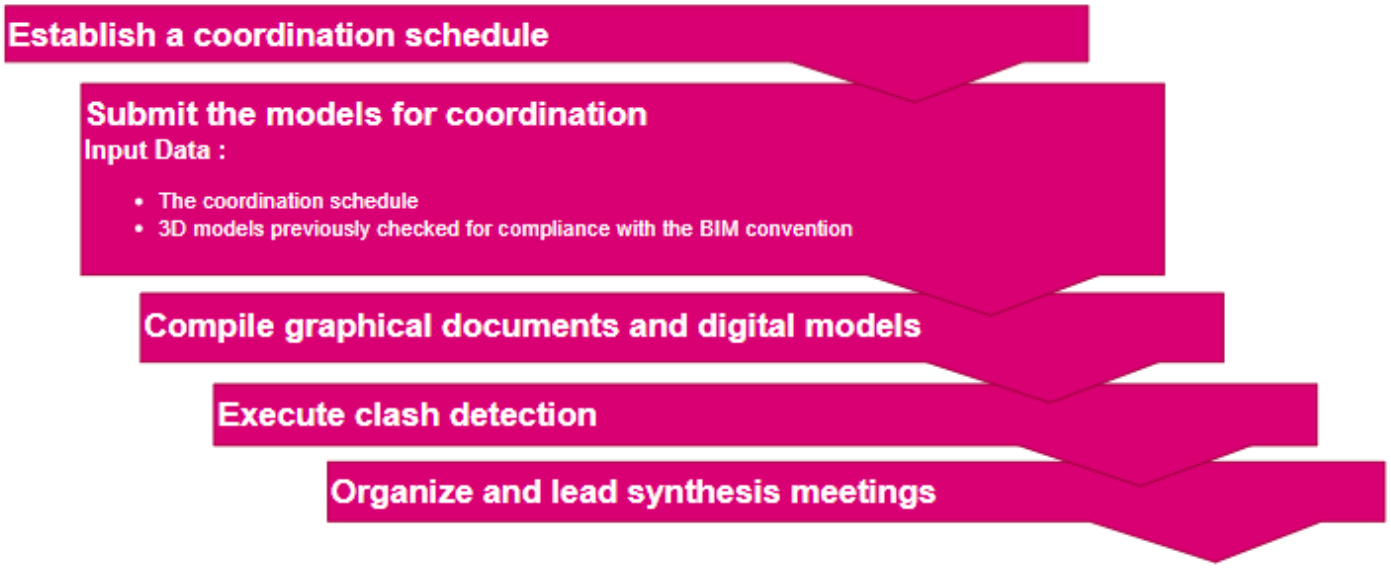
Benefits:

- Quickly highlight geometric clashes between different trades
- Generate detailed clash reports illustrated with 3D views
- Propose visually supported solutions based on 3D representations
- Reduce construction costs and project delays through early issue resolution

Skill requirements:

- Ability to manipulate, navigate, and review a 3D model
- Knowledge of BIM modelling solutions
- Understanding of construction systems
- Ability to work collaboratively and analyse multidisciplinary coordination issues

Procedure:



5.2.3 Model-Based Documentation

Description:

The process consists of producing drawings, diagrams, schedules, and quantity take-offs directly from the discipline models. This ensures consistency between documentation and the models and automates updates triggered by design changes.

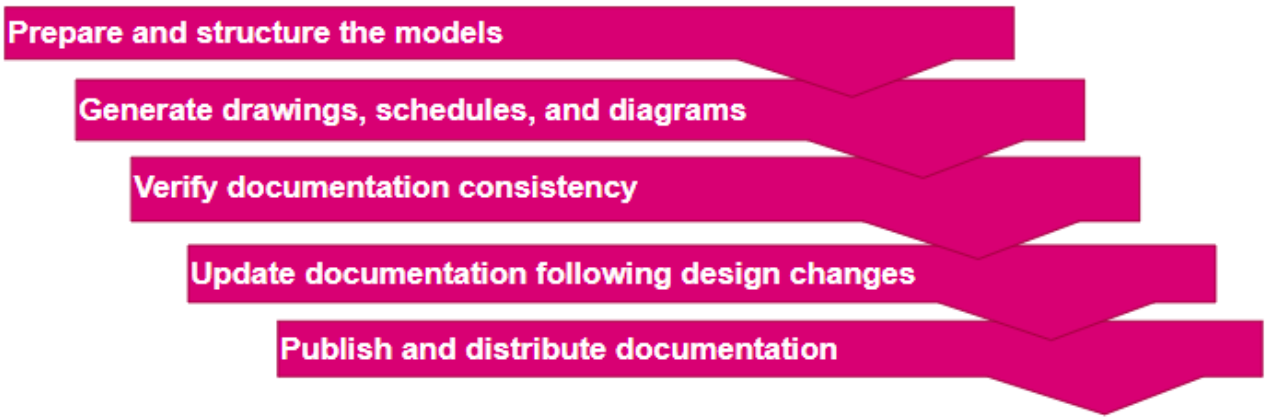
Benefits:

- Ensures perfect alignment between drawings and the 3D model
- Reduces errors related to manual updates or duplicated information
- Improves coordination between disciplines thanks to consistent datasets
- Facilitates rapid updates of drawings and schedules following design modifications
- Enhances the reliability of quantity take-offs and cost estimations

Skill requirements:

- Ability to generate and manipulate documentation directly from BIM models
- Knowledge of annotation standards and project documentation requirements
- Understanding of modelling best practices to ensure extractable data quality
- Ability to verify, validate, and format drawings according to project conventions

Procedure:



5.2.4 4D / Site Logistics Simulation

Description:

The process consists of visually sequencing construction activities, site organization, and phasing using time-based simulation tools. Although optional, this BIM use is highly recommended to enhance construction planning, anticipate constraints, and optimize site logistics.

Benefits:

- Provides a clear visual representation of the construction sequence
- Improves understanding of site constraints, access, and material flows
- Enhances coordination between trades by identifying planning conflicts early
- Facilitates communication with stakeholders through time-based visualizations
- Supports optimization of construction phasing, site layout, and resource allocation

Skill requirements:

- Ability to interpret and work with construction schedules (Gantt charts, phasing plans)
- Knowledge of 4D simulation and BIM coordination tools
- Understanding of construction sequencing, site logistics, and safety requirements
- Ability to analyse planning constraints and propose optimized phasing strategies

Procedure:

5.2.5 Asset Data Structuring

Description:

The process consists of integrating object-level information (attributes, classification, maintenance data) into the model to support future building operation, asset management, and data handover.

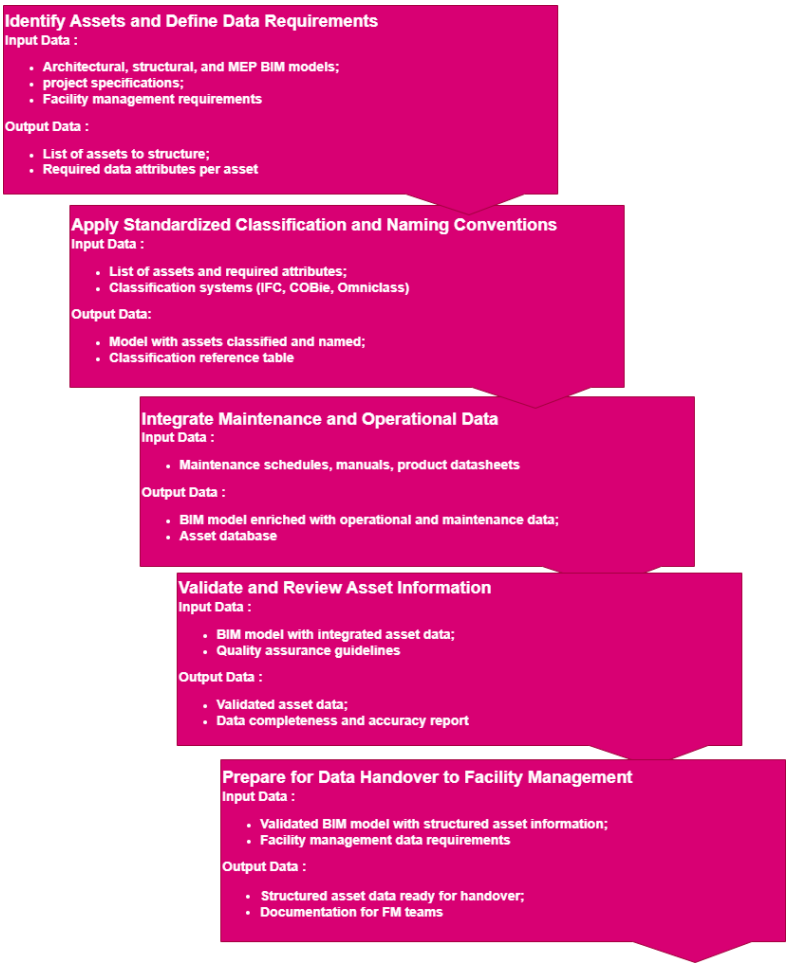
Benefits:

- Provides a structured and standardized dataset for all building elements
- Supports efficient operation, maintenance, and facility management
- Facilitates future upgrades, renovations, and retrofits by providing accurate information
- Enhances collaboration between design, construction, and operations teams
- Ensures smooth data handover to the client or facility management team

Skill requirements:

- Understanding of BIM standards and classification systems (e.g., IFC, COBie[8], Omniclass[9])
- Ability to manage and input detailed asset information in BIM software
- Knowledge of facility management and maintenance requirements
- Attention to detail and consistency for accurate data structuring
- Ability to verify and validate model data for completeness and usability

Procedure:



[8] *COBie is a standardized format for exporting BIM data to provide facility management teams with structured information about building assets, spaces, and maintenance requirements.*

[9] *OmniClass is a classification system that organizes all building elements and processes in a consistent way to ensure clarity and standardization across disciplines.*

5.2.6 Sustainability Analysis

Description:

The process consists of linking the BIM model to sustainability tools to evaluate the project's environmental performance. This includes assessing embodied carbon, estimating energy and water use, analyzing climate resilience, and preparing CSR indicators or certification outputs (LEED[10], EDGE[11]).
The objective is to support responsible design choices and help MKR reach its CSR goals.

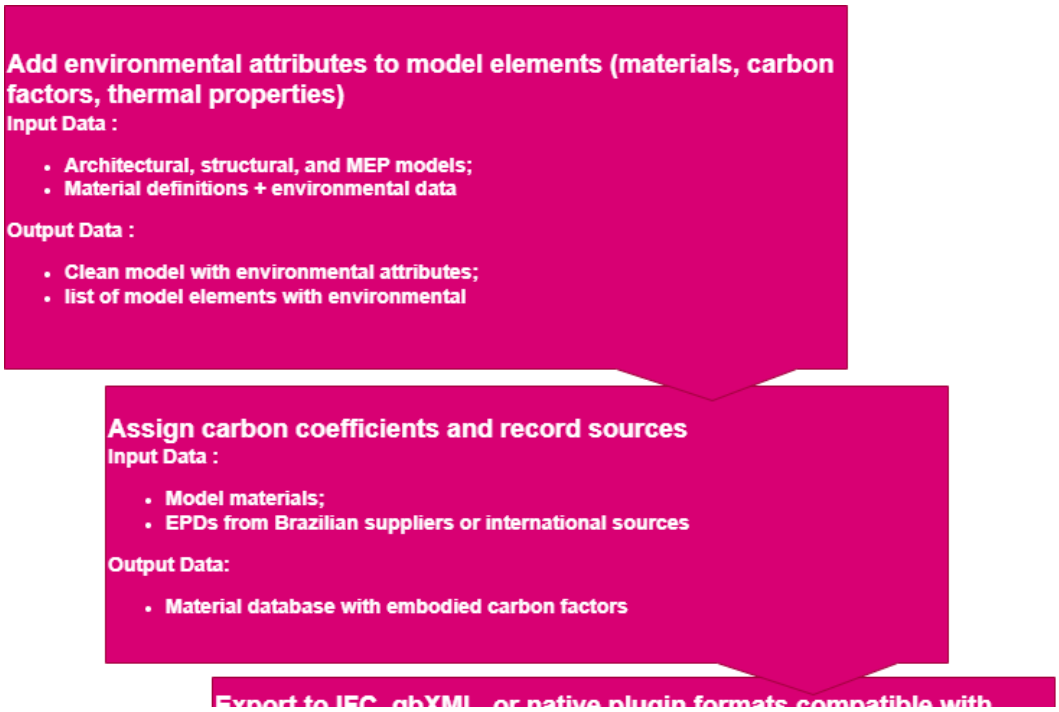
Benefits:

- Early identification of high-impact materials and potential optimizations.
- More accurate forecasting of operational energy consumption.
- Improved environmental reporting and CSR compliance.
- Ability to compare design options based on sustainability performance.
- Contribution to greener design choices and long-term efficiency.

Skill requirements:

- Early identification of high-impact materials
- Better prediction of energy and water performance
- Strong alignment with Brazilian CSR expectations
- Centralized sustainability data for the operation phase
- Easier achievement of environmental certifications
- Improved climate resilience and accessibility (NBR 9050[12])

Procedure:



[10] An international system that certifies buildings for overall sustainability and energy efficiency.
[11] A global certification focused on reducing energy, water, and material use in buildings.
[12] NBR 9050 is a Brazilian standard that establishes rules for accessibility in buildings, urban spaces, and facilities. It defines requirements for ramps, doors, bathrooms, signage, and circulation so that people with disabilities or reduced mobility can access and use spaces safely and comfortably.

Export to IFC, gbXML, or native plugin formats compatible with analysis tools
Input Data :
- Parameterized BIM model

Output Data :
- Valid analysis files for sustainability simulations

Run Embodied Carbon, Energy, and Water Simulations
Input Data :
- Exported analysis files;
- local climate and occupancy data

Output Data :
- Embodied carbon results;
- energy and water performance models

Analyze risks (overheating, rainfall, humidity) and accessibility compliance
Input Data :
- Simulation-ready model;
- Brazilian climate data + NBR 9050

Output Data :
- Climate resilience and accessibility diagrams

Prepare simulations/exports for certification pathways
Input Data :
- Analysis files + certification templates

Output Data :
- Simulations for sustainability certifications

Organize key performance indicators (energy, water, carbon, comfort)
Input Data :
- All sustainability outputs; CSR goals for the building

Output Data :
- CSR KPI dashboard proposals for operation

5.2.7 5D Cost Simulation

Description:

The process consists of integrating quantity take-offs and cost data with the model to simulate budgets, track cost evolution, and support financial decision-making. The model is linked to unit prices and updated automatically when design changes occur.

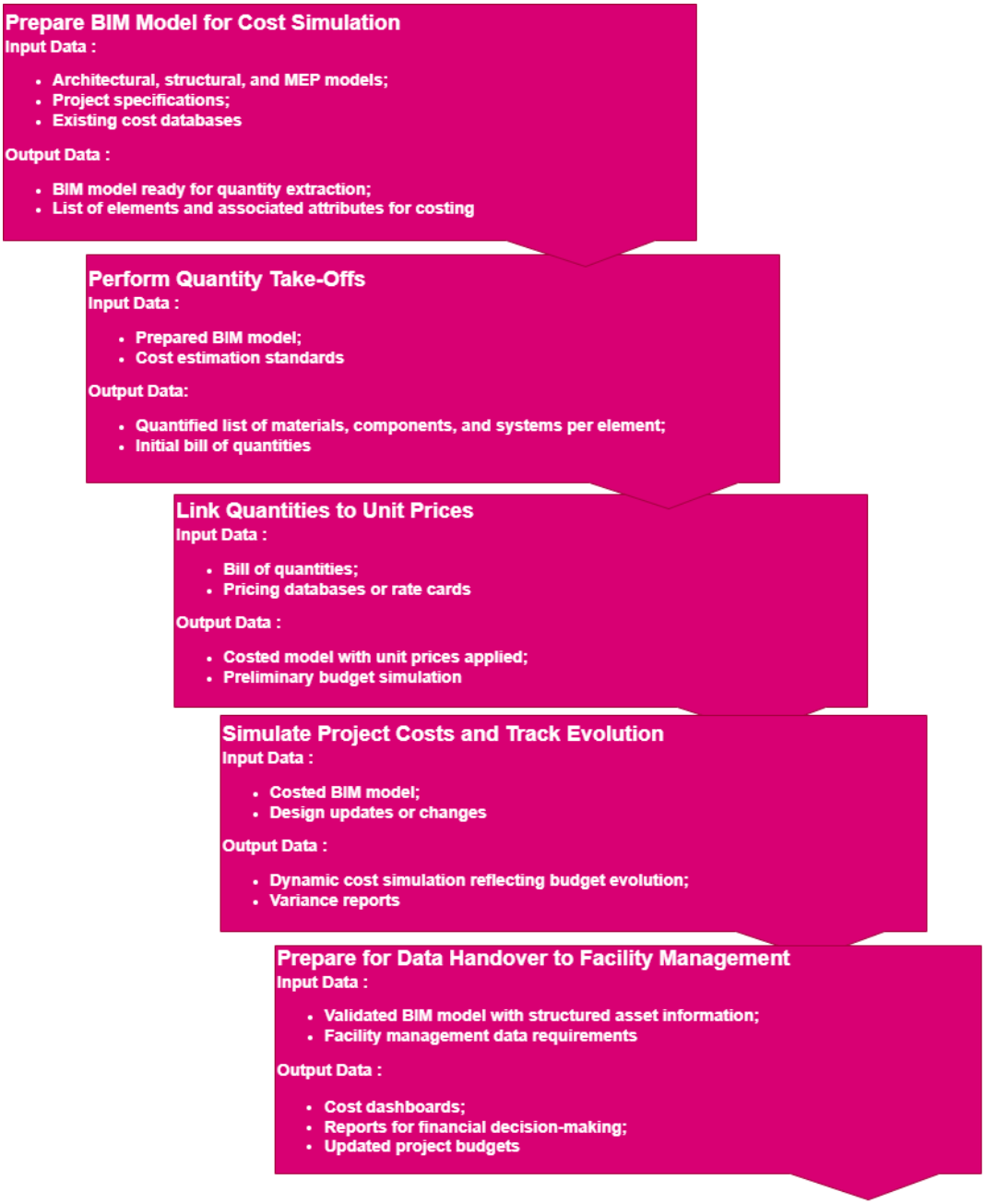
Benefits:

- Provides accurate and up-to-date cost estimates linked to the BIM model
- Allows early detection of budget deviations due to design changes
- Supports scenario analysis and financial decision-making
- Enhances collaboration between design, construction, and cost management teams
- Improves transparency and accountability in project budgeting

Skill requirements:

- Knowledge of quantity take-off methods and cost estimation
- Understanding of BIM modelling and linking models to cost databases
- Ability to analyse cost data and track budget evolution
- Familiarity with 5D simulation software and BIM-integrated cost tools
- Attention to detail to ensure correct mapping of quantities and unit prices

Procedure:



5.2.8 6D Facility Management Integration

Description:

The process consists of enriching the model with asset information required for building operation. This includes equipment data, maintenance schedules, warranties, and classification systems to ensure a structured and interoperable digital handover compatible with facility management tools.

Benefits:

- Provides a complete, structured dataset for building operation and maintenance
- Facilitates long-term asset management and lifecycle planning

- Improves efficiency of preventive maintenance and operational scheduling
- Ensures smooth handover to facility management teams with standardized information
- Supports integration with FM software for monitoring, reporting, and analysis

Skill requirements:

- Knowledge of facility management processes and maintenance planning
- Understanding of BIM standards and classification systems (COBie, OmniClass, IFC)
- Ability to manage and input detailed asset and operational data
- Familiarity with FM software and data interoperability requirements
- Attention to detail to ensure completeness and accuracy of data

Procedure:

5.2.9 7D Lifecycle & Decommissioning Analysis

Description:

The process consists of enriching the model with asset information required for building operation. This includes equipment data, maintenance schedules, warranties, and classification systems to ensure a structured and interoperable digital handover compatible with facility management tools.

Benefits:

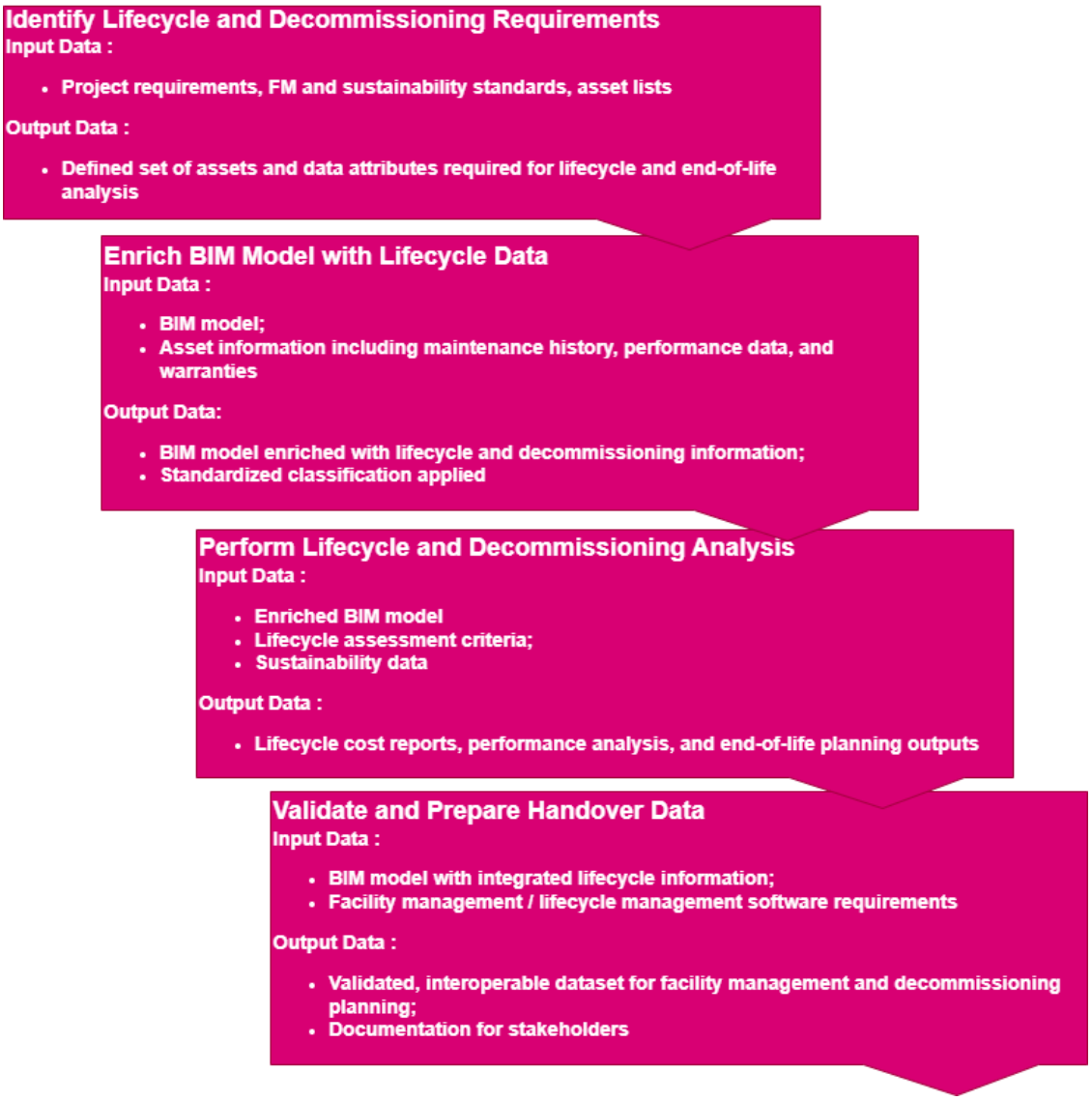
- Supports informed decision-making across the building lifecycle, from operation to decommissioning
- Facilitates lifecycle cost analysis and sustainability evaluation
- Enhance planning for refurbishment, reuse, or disposal of building assets
- Provides structured data to improve resource efficiency and environmental performance

- Ensures smooth handover of comprehensive lifecycle information to facility management and stakeholders

Skill requirements:

- Knowledge of building lifecycle management and decommissioning strategies
- Understanding of BIM standards and classification systems (COBie, OmniClass, IFC)
- Ability to manage and input operational, performance, and end-of-life data
- Familiarity with Facility management and lifecycle management software
- Analytical skills to assess asset performance and lifecycle costs

Procedure:



CHAPTER 6 Classification/Coding Rules

6.1 Object classification

All structural elements must follow a consistent naming convention to facilitate identification, tracking, and coordination in federated models.

- Naming Convention Components:
  - Project code: VBP
  - Discipline code: S = Structure
  - Element type / zone identifier: e.g., COL (Column), SLB (Slab), WL (Wall)
  - Unique element number: sequential identifier per type
  - Revision index: R01, R02, etc.

  - Status: WIP (Work-in-progress), SHARED, PUBLISHED, or ARCHIVED
- Example:
  EPB-S-COL-001-R01-SHARED.rvt
- Metadata Requirements:
  Each structural object must embed metadata to support coordination, approval, and project management:
  - Author / Modeling team
  - Date of creation / modification
  - Status (WIP / Shared / Published / Archived)
  - Version number
  - Approval status
  - Model purpose (e.g., Quantity Take-off, Structural Analysis, Construction Coordination)
  - Custom parameters (e.g., CSR indicators, material class) consistent across all structural models

6.2 Model file naming convention

All BIM models must follow the standardized naming structure below:

VBP_PHS_ENT_LOT_RXX_STATUS_YYYYMMDD

The separator is the underscore "_"

6.2.1 Naming Convention Components

Component	Description	Format	Example
VBP	Project Code (ElectroPark Brazil)	Fixed 3 letters	VBP
PHS	Phase Code	3 letters	DET, EXE, ASB
ENT	Entity/Contributor Code	3 letters	BEC, CES, MKR
LOT	Discipline/Trade Code	3 letters	STR, ARC, ELE
RXX	Revision Number	R + 2 digits	R01, R02, R15
STATUS	Model Status	3 letters	WIP, SHR, PUB, ARC
YYYYMMDD	Date Stamp	8 digits	20260315

6.2.2 Phase Codes (PHS)

Code	Phase Name	Description	Typical Duration
PRE	Preliminary Studies	Initial concept, feasibility, site analysis	2-3 months
SCH	Schematic Design	Preliminary design, massing, zoning	2-3 months
DET	Detailed Design	Full technical design, coordination	3-4 months
CON	Construction Documentation	Permit drawings, specifications	2 months
EXE	Execution/Construction	Construction phase modeling	12-18 months
ASB	As-Built	Final documentation, FM handover	1-2 months

6.2.3 Entity Codes (ENT)

Code	Entity Name	Role	Primary Responsibility
BEC	BEC Construction	Prime Contractor & Design	All disciplines design & construction
CES	CES International	AMOA/BIM Management	BIM coordination, quality control
MKR	MKR Corp	Client	Reference models, requirements
SUB	Subcontractors	Specialized trades	Specific systems (to be detailed)
EXT	External Consultants	Specialized studies	CSR, energy, geotechnical

Note: Subcontractors must use specific 3-letter codes assigned during mobilization (e.g., ELS for electrical subcontractor, MEP for MEP specialist).

6.2.4 Discipline/Trade Codes (LOT)

Code	Discipline Name	Scope	Typical Elements	Responsible Entity
REF	Reference Model	Site context, grids, levels	Base geometry, coordinates	CES/MKR
ARC	Architecture	Buildings, layout, circulation	Walls, floors, spaces, finishes	BEC
STR	Structure	Structural systems	Foundations, slabs, columns, beams	BEC
HVC	HVAC	Climate control systems	Ducts, equipment, controls	BEC/SUB
ELE	Electrical	Power distribution, lighting	Panels, circuits, fixtures, charging stations	BEC/SUB
PVS	Photovoltaic Systems	Solar energy production	PV panels, inverters, batteries, monitoring	BEC/EXT
PLB	Plumbing	Water systems	Pipes, fixtures, drainage	BEC/SUB
CWK	Civil Works	External infrastructure	Roads, drainage, landscaping	BEC
DIG	Digital Systems	IT & monitoring	Access control, CCTV, data networks	SUB/EXT
FPS	Fire Protection	Safety systems	Sprinklers, alarms, emergency systems	SUB
ENV	Environmental	Sustainability features	Rainwater, natural ventilation	EXT

6.2.5 Revision Index (RXX)

Format	Description	When to Increment
R01	First official revision	Initial submission to Shared
R02	Second revision	After coordination comments
R03+	Subsequent revisions	Each new submission cycle

R99	Final approved revision	Use only for final Published state

Rules:

- Revision numbers are sequential and never reset
- Internal WIP versions do not require revision numbers
- Revision log must be maintained in model properties
- Major design changes may skip numbers (e.g., R01 → R05) with justification

6.2.6 Model Status Codes (STATUS)

Code	Status Name	Meaning	CDE Location	Who Can Set
WIP	Work in Progress	Active development, not reviewed	WIP folder	Modeler
SHR	Shared	Ready for coordination	Shared folder	Coordinator
PUB	Published	Approved for use	Published folder	BIM Manager
ARC	Archived	Superseded version	Archive folder	CDE Manager
HLD	On Hold	Awaiting information	Shared folder	BIM Manager
REJ	Rejected	Does not meet requirements	Returned to WIP	BIM Manager

6.2.7 Date Stamp (YYYYMMDD)

The date stamp indicates the submission or last modification date.
Format: YYYYMMDD

- YYYY: 4-digit year (e.g., 2026)
- MM: 2-digit month (01-12)
- DD: 2-digit day (01-31)

Example: 20260315 = March 15, 2026
Purpose:

- Enables chronological sorting
- Supports version tracking
- Facilitates audit trails
- Required for all Shared, Published, and Archived files

6.2.8 Complete Naming Examples

Example Filename	Breakdown	Meaning
VBP_PRE_BEC_ARC_R01_SHR_20260115.rvt	VBP / PRE / BEC / ARC / R01 / SHR / 20260115	ElectroPark / Preliminary / BEC / Architecture / Rev 1 / Shared / Jan 15, 2026
VBP_DET_BEC_STR_R05_PUB_20260420.rvt	VBP / DET / BEC / STR / R05 / PUB / 20260420	ElectroPark / Detailed Design / BEC / Structure / Rev 5 / Published / Apr 20, 2026
VBP_EXE_SUB_ELE_R12_SHR_20261208.rvt	VBP / EXE / SUB / ELE / R12 / SHR / 20261208	ElectroPark / Execution / Subcontractor / Electrical / Rev 12 / Shared / Dec 8, 2026
VBP_ASB_BEC_PVS_R99_PUB_20270310.rvt	VBP / ASB / BEC / PVS / R99 / PUB / 20270310	ElectroPark / As-Built / BEC / Photovoltaic / Final / Published / Mar 10, 2027

VBP_DET_CES_REF_R02_PUB_20260201.rvt	VBP / DET / CES / REF / R02 / PUB / 20260201	ElectroPark / Detailed Design / CES / Reference / Rev 2 / Published / Feb 1, 2026
VBP_CON_BEC_CVC_R08_HLD_20260715.rvt	VBP / CON / BEC / CVC / R08 / HLD / 20260715	ElectroPark / Construction Doc / BEC / HVAC / Rev 8 / On Hold / Jul 15, 2026

6.2.9 Linked Files and References

For files that contain linked models (e.g., federated coordination models):
Format: VBP_PHS_FED_ALL_RXX_STATUS_YYYYMMDD.nwf

Component	Description
FED	Indicates federated/coordination model
ALL	Contains all disciplines

Example:

- VBP_DET_FED_ALL_R03_SHR_20260420.nwf - Federated coordination model, detailed design, revision 3

6.2.10 Special Model Types

Model Type	Code	Example	Purpose
Coordination Model	FED	VBP_DET_FED_ALL_R02_SHR_20260320	Combined disciplines for clash detection
Site Survey	SUR	VBP_PRE_MKR_SUR_R01_PUB_20260105	Existing conditions, topography
4D Sequence Model	4DM	VBP_EXE_BEC_4DM_R01_SHR_20260915	Construction phasing simulation
Energy Analysis	ENR	VBP_DET_EXT_ENR_R03_SHR_20260510	Energy performance modeling
Cost Estimation	5DM	VBP_CON_BEC_5DM_R02_SHR_20260820	Quantity takeoffs and cost analysis

6.2.11 File Naming Validation Checklist

Before uploading any model to the CDE, verify:

- All 7 components present in correct order
- Project code is **VBP** (uppercase)
- Phase code matches current project phase
- Entity code is correct and authorized
- Discipline code matches model content
- Revision number is sequential and correct
- Status code reflects current approval state
- Date stamp is accurate (YYYYMMDD format)
- No spaces, special characters, or accents
- File extension matches software (.rvt, .nwf, .ifc, etc.)

Non-compliance results in:

- Automatic rejection by CDE validation script
- Return to author for correction
- Delay in coordination cycle
- Potential impact on project schedule

6.2.12 Automated File Naming Tools

To ensure consistency and reduce errors, the following tools are recommended:

Revit Plugin:
- File Naming Automation Tool (to be provided by CES)
- Auto-populates file properties with correct naming components
- Validates naming before save

CDE Validation:
- Autodesk Construction Cloud has built-in naming validation
- Upload will be rejected if naming convention is not followed
- Error messages guide users to correct format

Excel Template:
- File naming generator spreadsheet (available in CDE)
- Input parameters, receive correctly formatted filename
- Includes validation rules and examples

6.3 Object and family naming convention

All BIM objects and families must be coded and deposited on the platform following these principles:

ENT_LOT_CAT_SCAT_Description_usg

This codification is organized around 6 terms. The terms CAT, SCAT, and usg are optional and used as needed.
For object denomination, use only non-accented, non-subscripted alphabet letters.
The separator is the underscore "_"

ENT - Entity Code

3-letter code of the contributor:
- **BEC**: BEC Construction
- **CES**: CES International
- **MKR**: MKR Corp

LOT - Discipline Code

3-letter designation of the relevant discipline:
- **ARC**: Architecture and site layout
- **STR**: Structure
- **HVC**: HVAC
- **ELE**: Electrical
- **PVS**: Photovoltaic systems
- **DIG**: Digital systems

CAT - Category

3-letter family category code.

Examples:
- **COL**: Columns
- **SLB**: Slabs
- **CHG**: Charging stations
- **PNL**: Solar panels
- **FOU**: Foundations

SCAT - Sub-category

4-letter sub-category code (optional).

Examples:
- **FAST**: Fast charging
- **SLOW**: Slow charging
- **ROOF**: Roof-mounted
- **CANO**: Canopy-mounted

"This term is optional and provides more precision in object classification"

Description

Term corresponding to the object name

usg - Usage

Term determining object usage (specific to Revit families):
- **imb**: Nested family in a project family
- **sym**: Symbol family
- Leave blank for project families

Examples:

- BEC_ELE_CHG_FAST_Type2_50kW - Fast charging station, Type 2, 50kW
- BEC_PVS_PNL_ROOF_Bifacial_550W - Roof-mounted bifacial solar panel, 550W
- BEC_STR_COL_Rectangular_400x400 - Rectangular structural column, 400x400mm
- BEC_ARC_PARK_Standard_Vehicle_Space - Standard vehicle parking space

6.4 View naming convention (Revit-specific)

Models from software other than Revit must at minimum respect the following conditions:

- Have clear and concise codification
- Organize views correctly within models
- Manage views by levels

For Revit models, the view tree must respect the following organization:

LOT_TYP_View name

The separator is the underscore "_". Use only non-accented, non-subscripted alphabet letters.

This organization is structured around 2 terms that allow sorting of different views by discipline and type.

LOT - Discipline Code

3-letter designation of relevant discipline:

- **ARC**: Architecture
- **STR**: Structure
- **CVC**: HVAC
- **ELE**: Electrical
- **PVS**: Photovoltaic
- **DIG**: Digital systems

TYP - View Type

3-letter designation:

- **PLN**: Floor plan
- **SEC**: Section
- **DET**: Detail
- **ELV**: Elevation
- **3DV**: 3D view
- **SCH**: Schedule/Legend
- **RCP**: Reflected ceiling plan
- **SHT**: Sheet

Examples:

- STR_PLN_Level_VS_Foundation
- ELE_SEC_Charging_Station_Detail
- PVS_3DV_Solar_Array_Overview
- ARC_PLN_Level_02_Parking

CHAPTER 7 COMMON DATA ENVIRONMENT (CDE)

7.1 CDE Platform Selection

For the ElectroPark Brazil project, the approved Common Data Environment is:

Autodesk Construction Cloud (ACC)

This platform has been selected for its:

- Native integration with Autodesk Revit and Navisworks
- Robust version control and access management
- Compliance with ISO 19650 workflows
- Support for IFC 4.3 and multiple file formats
- Real-time collaboration and mobile access
- Integration with Brazilian data protection requirements

7.2 CDE Structure and Access Rights

7.2.1 Folder Structure

The CDE follows the ISO 19650 information container structure with four main states:

1. WORK IN PROGRESS (WIP)

- Individual or team working files

- Not visible to other disciplines
- No formal naming requirements until sharing

2. SHARED

- Files shared for coordination and review
- Visible to all authorized project members
- Subject to naming conventions
- Requires completed submission checklist

3. PUBLISHED

- Approved and authorized information
- Formally approved for use
- Immutable (no further edits)
- Official project reference

4. ARCHIVE

- Superseded information
- Retained for record purposes
- Read-only access

7.2.2 Detailed Folder Architecture

```
VBP_ElectroPark_Brazil/
|
 ├── 00_Project_Management/
|    ├── 00.1_Contracts_Legal/
|    ├── 00.2_BIM_Execution_Plans/
|    ├── 00.3_Meeting_Minutes/
|    ├── 00.4_Project_Directory/
|    └── 00.5_Quality_Control/
|
 ├── 01_Reference_Information/
|    ├── 01.1_Site_Surveys/
|    ├── 01.2_Geotechnical_Reports/
|    ├── 01.3_Existing_Conditions/
|    ├── 01.4_Regulatory_Documents/
|    └── 01.5_Standards_Templates/
|
 ├── 02_Design_Development/
|    |
|    ├── 02.1_Architecture/
|    |    ├── WIP/
|    |    ├── Shared/
|    |    ├── Published/
|    |    └── Archive/
|    |
|    ├── 02.2_Structure/
|    |    ├── WIP/
|    |    ├── Shared/
|    |    ├── Published/
|    |    └── Archive/
|    |
|    ├── 02.3_MEP_Systems/
|    |    ├── WIP/
|    |    ├── Shared/
|    |    ├── Published/
|    |    └── Archive/
|    |
```

```
│   ├── 02.4_Photovoltaic/
│   │   ├── WIP/
│   │   ├── Shared/
│   │   ├── Published/
│   │   └── Archive/
│   │
│   ├── 02.5_Digital_Systems/
│   │   ├── WIP/
│   │   ├── Shared/
│   │   ├── Published/
│   │   └── Archive/
│   │
│   └── 02.6_Site_Works/
│       ├── WIP/
│       ├── Shared/
│       ├── Published/
│       └── Archive/
│
├── 03_Coordination/
│   ├── 03.1_Federated_Models/
│   ├── 03.2_Clash_Detection/
│   │   ├── Reports/
│   │   └── BCF_Files/
│   ├── 03.3_Coordination_Meetings/
│   ├── 03.4_Issue_Logs/
│   └── 03.5_Model_Validation/
│
├── 04_Documentation/
│   ├── 04.1_Drawings/
│   ├── 04.2_Specifications/
│   ├── 04.3_Schedules/
│   └── 04.4_Quantity_Takeoffs/
│
├── 05_Sustainability_CSR/
│   ├── 05.1_Environmental_Analysis/
│   ├── 05.2_Energy_Modeling/
│   ├── 05.3_Material_Declarations/
│   ├── 05.4_CSR_Reports/
│   └── 05.5_Certification_Documents/
│
├── 06_Construction/
│   ├── 06.1_Construction_Models/
│   ├── 06.2_Shop_Drawings/
│   ├── 06.3_Submittals/
│   ├── 06.4_RFIs/
│   └── 06.5_Site_Documentation/
│
└── 07_Handover/
    ├── 07.1_AsBuilt_Models/
    ├── 07.2_COBie_Data/
    ├── 07.3_O&M_Manuals/
    ├── 07.4_Warranties/
    └── 07.5_FM_Integration/
```

7.2.3 Access Rights Matrix

Role	WIP	Shared	Published	Archive	Project Management	CSR Folder	Coordination
MKR BIM Representative	View only	R/W/C	R/W/A	View only	R/W	R/W	View all
AMOA BIM Manager	View all	R/W/C	R/W/A	R/W	R/W	R/W/C	R/W
CDE Manager	R/W/D	R/W/D	R/W/D	R/W/D	R/W/D	R/W	R/W/D
BIM Coordinator	R/W (own)	R/W (own) / View (others)	View all	View all	View only	R (if relevant)	R/W
BIM Modeler	R/W (own files)	View only	View only	View only	No access	No access	View only
CSR Integrator	View as needed	R/W (CSR data)	View all	View all	View only	R/W	View only
Prime Contractor	View as needed	View all	View all	View all	R/C	View only	R/W
External Consultants	No access	View (authorized)	View (authorized)	No access	No access	No access	View only

Legend:

- **R**: Read
- **W**: Write/Edit
- **C**: Comment/Annotate
- **D**: Delete/Archive
- **A**: Approve/Authorize
- **R/W**: Read and Write
- **R/W/C**: Read, Write and Comment
- **R/W/D**: Read, Write and Delete
- **R/W/A**: Read, Write and Approve

Notes:

- All permissions are subject to role-based authentication and logging
- "Own" refers to files created by or assigned to the user's discipline/team
- "View all" includes read-only access to all project information within that state
- External consultants require explicit authorization per folder
- Prime Contractor has elevated access during construction phase only

7.3 File Exchange Protocols

7.3.1 Supported Formats

File Type	Purpose	Mandatory Format	Optional Format
BIM Models	Native authoring	.rvt, .pln, .dwg	-
Exchange Models	Coordination	IFC 4.3	IFC 2x3
Coordination	Federated models	.nwf, .nwd	-
Drawings	Documentation	.pdf	.dwg, .dxf
Schedules	Data extraction	.xlsx	.csv
Issues	Clash tracking	.bcf	.pdf
Documents	Reports, specs	.pdf	.docx
Images	Renderings	.png, .jpg	-

7.3.2 File Size Limitations

- Maximum file size for upload: **500 MB**
- Models exceeding this limit must be split logically
- Large files may be shared via dedicated transfer protocols with CDE Manager approval

7.3.3 Version Control

All files in the Shared and Published states must follow version control rules:
Format: Filename_RXX_STATUS_YYYYMMDD
Example: VBP_DET_BEC_STR_R03_SHR_20260315.rvt

- Revision numbers increment with each submission
- Date stamp indicates submission date
- Previous versions automatically moved to Archive by CDE Manager

7.4 Submission and Validation Process

7.4.1 Submission Checklist

Before any file is moved from WIP to Shared, the BIM Coordinator must complete a digital submission checklist including:

1. **File Integrity**
    - File opens without errors
    - All links and references are valid
    - No corrupted elements
2. **Naming Compliance**
    - File name follows convention
    - All objects properly coded
    - Views correctly organized
3. **Model Quality**
    - Target LOD achieved
    - All required metadata present
    - No critical warnings
4. **Coordination Readiness**
    - Model properly geo-located
    - Levels and grids aligned with reference model
    - No duplicate geometry
5. **Documentation**
    - Accompanying technical notes provided
    - Known issues flagged
    - Change log updated

7.4.2 Approval Workflow

[BIM Modeler]
↓ Completes work
[BIM Coordinator Review]
↓ Internal discipline check
[SHARED State]
↓ Upload to CDE
[Interdisciplinary Review]
↓ Coordination check
[AMOA BIM Manager Validation]
↓ Quality control
[Client Approval] (if required)
↓ Final authorization
[PUBLISHED State]
↓ Official reference
[Previous version → ARCHIVE]

7.4.3 Review Timelines

Submission Type	Review Period	Approver
WIP to Shared	2 business days	BIM Coordinator

Shared coordination	5 business days	All disciplines
BIM Manager validation	3 business days	AMOA BIM Manager
Client approval	7 business days	MKR Representative

7.5 Model Federation and Coordination

7.5.1 Federation Schedule

Federated models will be created at the following intervals:

- **Weekly**: During active design phases
- **Bi-weekly**: During documentation phase
- **Before major milestones**: As defined in project schedule
- **On-demand**: Upon request for specific coordination sessions

7.5.2 Federation Process

1. **Collect Models**: CDE Manager retrieves all current Shared models
2. **Validate Links**: Verify all models load correctly in federation tool
3. **Run Clash Detection**: Execute automated clash detection routines
4. **Generate Report**: Create clash report with BCF file
5. **Distribute**: Share federated model and report via CDE
6. **Coordination Meeting**: Review findings in scheduled session

7.5.3 Clash Detection Matrix

Discipline A	Discipline B	Clash Type	Priority	Review Frequency
Architecture	**Structure**	Hard	Critical	Weekly
Structure	**MEP**	Hard	Critical	Weekly
MEP	**PV Systems**	Hard	High	Bi-weekly
Architecture	**Digital Systems**	Soft	Medium	Bi-weekly
All	**Site Works**	Hard	High	Weekly

Clash Priority Levels:

- **Critical**: Must be resolved before progression
- **High**: Should be resolved within current phase
- **Medium**: Can be flagged for future resolution
- **Low**: Advisory only

7.6 Information Security and Data Protection

7.6.1 Data Protection Compliance

All CDE operations must comply with:

- Brazilian General Data Protection Law (LGPD)
- ISO 27001 information security standards
- MKR Corp internal IT security policies

7.6.2 Access Security

- All users require unique login credentials
- Two-factor authentication mandatory for external users
- Session timeout after 30 minutes of inactivity
- All access logged and auditable
- Regular access rights reviews (quarterly)

7.6.3 Data Backup

- Automated daily backups of entire CDE
- Weekly full system snapshots

- 90-day retention policy for deleted files
- Disaster recovery plan in place
- Backup storage on geographically separate servers

7.7 Training and Support

7.7.1 CDE Training Requirements

All project participants must complete:

- **Initial CDE orientation** (2 hours) - Before project start
- **Role-specific training** (2-4 hours) - Within first week
- **Refresher sessions** (1 hour) - Quarterly

Training covers:

- Navigation and file management
- Upload/download procedures
- Version control
- Issue logging and BCF workflows
- Security and data protection

7.7.2 Technical Support

Support Level	Response Time	Contact Method
Critical (CDE down)	1 hour	Phone: +55 (XX) XXXX-XXXX
High (Cannot upload/access)	4 hours	Email: cde.support@CESI.com
Medium (Process questions)	1 business day	Support ticket system
Low (General inquiries)	2 business days	Email

The CDE Manager provides first-line support with escalation to Autodesk technical support as needed.

7.8 CDE Roles and Responsibilities

7.8.1 CDE Manager Responsibilities

- Maintain folder structure and access rights
- Monitor storage capacity and performance
- Coordinate version control and archiving
- Generate usage reports and statistics
- Provide user training and support
- Enforce naming conventions
- Manage backup and recovery procedures
- Liaise with software vendor for technical issues

7.8.2 User Responsibilities

All CDE users must:

- Follow naming conventions strictly
- Complete submission checklists before uploads
- Respect folder structure and file organization
- Never delete or modify files outside their authority
- Report technical issues promptly
- Maintain confidentiality of login credentials
- Participate in required training sessions
- Review and acknowledge CDE usage policies annually

CHAPTER 8 SOFTWARE ENVIRONMENT

8.1 Approved Software Solutions

8.1.1 Design Phase Software

Discipline	Primary Software	Version
Architecture	Autodesk Revit	2025
Structure	Autodesk Revit	2025
MEP	Revit MEP	2025
Photovoltaic	PVsyst / Helioscope	Latest
Energy Analysis	IES VE	Latest

8.1.2 Coordination and Review

Function	Software	Version
Model Federation	Navisworks Manage	2025
Clash Detection	Solibri Office	Latest
Issue Management	BIMcollab	Latest
IFC Viewing	BIM Vision	Latest

8.1.3 Sustainability and CSR

Analysis Type	Software
Embodied Carbon	One Click LCA
Energy Simulation	IES VE / DesignBuilder
Daylighting	DIALux
Water Analysis	Autodesk Insight

8.1.4 Documentation and Collaboration

Purpose	Software
CDE Platform	Autodesk Construction Cloud
Document Management	ACC Docs
Mobile Coordination	ACC Mobile App
Data Extraction	Microsoft Excel / COBie Plugin

8.2 Software Interoperability Requirements

- All native models must export to **IFC 4.3** without data loss
- IFC exports must be validated before submission
- Coordinate systems must align across all software platforms
- Shared parameters and classification systems must be consistent

ANNEX A - STRUCTURAL LOD MATRIX

Level of Development (LOD) by Structural Element and Project Phase

As **Structural BIM Coordinator** for ElectroPark Brazil, the following LOD matrix defines the required geometric detail, information content, and documentation accuracy for all structural elements across project phases.

This matrix is based on:

- LOD Specification 2016 (BIMForum)
- ISO 19650 Level of Information Need principles
- MKR Corp BIM Requirements
- Brazilian structural standards (ABNT NBR series)

A.1 LOD Definition Summary

LOD	Geometry	Information	Use	Phase Alignment
100	Symbolic representation	Conceptual data	Volume, massing, feasibility	Preliminary Studies
200	Approximate geometry	Generic systems	Spatial coordination, preliminary design	Schematic Design
300	Precise geometry	Specific assemblies	Detailed coordination, quantities	Detailed Design
350	Precise with interfaces	Detailed coordination data	Construction coordination, fabrication prep	Construction Documentation
400	Fabrication detail	Complete fabrication data	Construction, shop drawings	Execution
500	As-constructed	Verified field conditions	Operations, maintenance, FM	As-Built

A.2 Structural Elements LOD Matrix

A.2.1 Foundations

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Spread Footings	Yes	Yes	Yes	Yes	Yes	Yes
Pile Foundations	Yes	Yes	Yes	Yes	Yes	Yes
Pile Caps	Yes	Yes	Yes	Yes	Yes	Yes
Foundation Beams (Grade Beams)	Yes	Yes	Yes	Yes	Yes	Yes
Foundation Slabs (SOG)	Yes	Yes	Yes	Yes	Yes	Yes
Retaining Walls (Below Grade)	No	Yes	Yes	Yes	Yes	Yes
Drainage Systems (Foundation)	No	Yes	Yes	Yes	Yes	Yes

LOD 100 (Preliminary):
- Symbolic representation only
- Approximate depth and bearing capacity from geotechnical report
- Conceptual foundation type (shallow vs deep)

LOD 200 (Schematic):
- Generic foundation elements with approximate dimensions
- Pile quantities and spacing (preliminary)
- Foundation depth based on soil report
- Generic concrete specification (e.g., C25/30)

LOD 300 (Detailed Design):
- Precise dimensions (length, width, thickness)
- Reinforcement design (main bars, stirrups)
- Concrete class (C25/30, C30/37)
- Load-bearing capacity
- Connection details to superstructure
- Embedments for MEP

LOD 350 (Construction Documentation):
- Complete reinforcement schedule
- Bar bending schedules
- Formwork requirements
- Construction joints
- Waterproofing details
- Interface with drainage systems

LOD 400 (Execution):
- Shop drawings for reinforcement
- As-ordered concrete specifications
- Installation sequence
- Tolerances and quality control parameters
- Manufacturer data for embedments

LOD 500 (As-Built):
- Verified installed dimensions
- Actual reinforcement as installed
- Concrete test results
- Foundation settlement monitoring data
- Scan data if available

A.2.2 Basement and Multi-Level Parking Slabs

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Basement Slab (VS Level)	Yes	Yes	Yes	Yes	Yes	Yes
Parking Level Slabs (L01-L04)	Yes	Yes	Yes	Yes	Yes	Yes
Ramps (Vehicle Circulation)	Yes	Yes	Yes	Yes	Yes	Yes
Post-Tensioned Slabs	No	Yes	Yes	Yes	Yes	Yes
Precast Slab Panels	No	Yes	Yes	Yes	Yes	Yes
Slab Edge Details	No	No	Yes	Yes	Yes	Yes
Openings (Ramps, Services)	No	Yes	Yes	Yes	Yes	Yes

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Precast Slab Panels	No	Yes	Yes	Yes	Yes	Yes
Slab Edge Details	No	No	Yes	Yes	Yes	Yes
Openings (Ramps, Services)	No	Yes	Yes	Yes	Yes	Yes

LOD 100 (Preliminary):
- Simple horizontal planes
- Level heights only
- Conceptual span lengths

LOD 200 (Schematic):
- Slab thickness (preliminary)
- Span directions/
- Support locations (columns/walls)
- Ramp slopes and widths
- Opening locations (approximate)

LOD 300 (Detailed Design):
- Precise slab thickness (e.g., 250mm, 300mm)
- Reinforcement design (top/bottom mesh)
- Drop panels or capitals if required
- Drainage slopes (1-2%)
- Vehicle load ratings (kN/m²)
- Concrete class and finish requirements
- MEP penetrations coordinated
- Expansion joints

LOD 350 (Construction Documentation):
- Complete reinforcement drawings
- Bar schedules with bending details
- Construction sequence and pour strips
- Formwork requirements
- Post-tensioning tendon layouts (if applicable)
- Interface details with walls and columns
- Waterstop locations

LOD 400 (Execution):
- Shop drawings for reinforcement
- Precast panel fabrication drawings (if applicable)
- Post-tensioning stressing sequence
- Concrete mix design
- Quality control checkpoints
- Surface finish specifications

LOD 500 (As-Built):
- Verified slab thickness and levels
- Actual reinforcement placement
- Concrete strength test results
- Deflection measurements
- Surface flatness (FF/FL numbers)
- Drainage performance verification

A.3.3 Columns

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Reinforced Concrete Columns	Yes	Yes	Yes	Yes	Yes	Yes
Precast Columns	No	Yes	Yes	Yes	Yes	Yes
Steel Columns (if any)	No	Yes	Yes	Yes	Yes	Yes
Column Capitals	No	No	Yes	Yes	Yes	Yes
Column Base Plates	No	No	Yes	Yes	Yes	Yes

LOD 100 (Preliminary):
- Generic vertical elements
- Approximate grid locations
- Conceptual sizing

LOD 200 (Schematic):
- Column dimensions (e.g., 400x400mm)
- Column spacing on grid
- Typical floor-to-floor heights
- Generic concrete strength

LOD 300 (Detailed Design):
- Precise cross-sections (rectangular, circular)
- Reinforcement design (longitudinal bars, ties)
- Concrete class (C30/37, C35/45)
- Column schedules with varying sections per level
- Load capacities
- Embedments for architectural/MEP

LOD 350 (Construction Documentation):
- Complete reinforcement details
- Bar schedules and bending details
- Splice locations and lap lengths
- Column base connection details
- Formwork requirements
- Construction tolerances

LOD 400 (Execution):
- Shop drawings for reinforcement cages
- Precast fabrication drawings (if applicable)
- Installation sequence
- Lifting points for precast
- Connection hardware specifications

LOD 500 (As-Built):
- Verified column dimensions and alignment
- Actual reinforcement as installed
- Verticality measurements
- Concrete strength records
- Connection test results

A.2.4 Beams and Girders

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Reinforced Concrete Beams	Yes	Yes	Yes	Yes	Yes	Yes
Transfer Beams	No	Yes	Yes	Yes	Yes	Yes

Edge Beams (Perimeter)	No	Yes	Yes	Yes	Yes	Yes
Precast Beams	No	Yes	Yes	Yes	Yes	Yes
Steel Beams (if any)	No	Yes	Yes	Yes	Yes	Yes

LOD 100 (Preliminary):
- Symbolic linear elements
- Approximate span lengths

LOD 200 (Schematic):
- Beam depths and widths (typical)
- Support locations
- Generic loading assumptions

LOD 300 (Detailed Design):
- Precise beam dimensions (width x depth)
- Reinforcement design (top/bottom bars, stirrups)
- Concrete class
- Beam schedules
- Connection details to columns and slabs
- Openings for MEP services

LOD 350 (Construction Documentation):
- Complete reinforcement drawings
- Bar schedules with bending details
- Stirrup spacing and configuration
- Splice locations
- Formwork and shoring requirements
- Interface with architectural elements

LOD 400 (Execution):
- Shop drawings for reinforcement
- Precast fabrication details (if applicable)
- Installation sequence
- Temporary support requirements
- Connection hardware

LOD 500 (As-Built):
- Verified beam dimensions
- Actual reinforcement placement
- Deflection measurements
- Concrete test results
- Connection integrity verification

A.2.5 Structural Walls

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Shear Walls	Yes	Yes	Yes	Yes	Yes	Yes
Core Walls (Stair/Elevator)	Yes	Yes	Yes	Yes	Yes	Yes
Retaining Walls (Above Grade)	No	Yes	Yes	Yes	Yes	Yes
Precast Wall Panels	No	Yes	Yes	Yes	Yes	Yes

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Shear Walls	Yes	Yes	Yes	Yes	Yes	Yes

Core Walls (Stair/Elevator)	Yes	Yes	Yes	Yes	Yes	Yes
Retaining Walls (Above Grade)	No	Yes	Yes	Yes	Yes	Yes
Precast Wall Panels	No	Yes	Yes	Yes	Yes	Yes

LOD 100 (Preliminary):
- Generic vertical surfaces
- Approximate locations

LOD 200 (Schematic):
- Wall thickness (preliminary)
- Wall lengths and heights
- Generic structural function

LOD 300 (Detailed Design):
- Precise wall thickness (e.g., 200mm, 250mm)
- Reinforcement design (vertical/horizontal)
- Concrete class
- Openings for doors, windows, MEP
- Connection to slabs and foundations
- Boundary element details (if required)

LOD 350 (Construction Documentation):
- Complete reinforcement details
- Bar schedules and bending details
- Construction joints and pour sequences
- Embedments and blockouts
- Formwork requirements
- Interface with architectural finishes

LOD 400 (Execution):
- Shop drawings for reinforcement
- Precast fabrication details (if applicable)
- Installation sequence
- Lifting and bracing details
- Connection hardware

LOD 500 (As-Built):
- Verified wall dimensions and alignment
- Actual reinforcement placement
- Concrete strength records
- Opening locations verified
- Waterproofing verification (if applicable)

A.2.6 Stairs and Ramps (Structural)

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Concrete Stairs	No	Yes	Yes	Yes	Yes	Yes
Precast Stair Flights	No	Yes	Yes	Yes	Yes	Yes
Landing Slabs	No	Yes	Yes	Yes	Yes	Yes
Structural Support for Stairs	No	No	Yes	Yes	Yes	Yes

LOD 200 (Schematic):
- Generic stair geometry
- Riser/tread dimensions (approximate)

- Landing locations

LOD 300 (Detailed Design):
- Precise tread/riser dimensions
- Stair thickness
- Reinforcement design
- Connection details to landings and floors
- Load ratings

LOD 350 (Construction Documentation):
- Complete reinforcement details
- Stair geometry for fabrication
- Formwork requirements
- Nosing and edge details

LOD 400 (Execution):
- Shop drawings for precast stairs
- Installation sequence
- Temporary support requirements
- Connection hardware

LOD 500 (As-Built):
- Verified dimensions
- Actual reinforcement
- Alignment and levelness verification

A.2.7 Roof Structure and PV Support

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Roof Slab	Yes	Yes	Yes	Yes	Yes	Yes
PV Panel Support Structure	No	Yes	Yes	Yes	Yes	Yes
Roof Beams	No	Yes	Yes	Yes	Yes	Yes
Parapets (Structural)	No	No	Yes	Yes	Yes	Yes

LOD 200 (Schematic):
- Roof slab thickness (preliminary)
- PV support concept (ballasted vs anchored)
- Drainage slopes

LOD 300 (Detailed Design):
- Precise roof slab design
- PV support structure dimensions and spacing
- Wind and seismic load considerations
- Attachment points to roof structure
- Waterproofing interface

LOD 350 (Construction Documentation):
- Complete structural details for PV supports
- Anchor bolt layouts
- Connection details
- Penetration/waterproofing coordination

LOD 400 (Execution):
- Shop drawings for PV support fabrication
- Installation sequence
- Load testing requirements
- As-ordered specifications

LOD 500 (As-Built):
- Verified PV support locations
- Load test results

- Roof penetration inspection records

A.2.8 Connections and Joints

Element Type	PRE (LOD 100)	SCH (LOD 200)	DET (LOD 300)	CON (LOD 350)	EXE (LOD 400)	ASB (LOD 500)
Column-Beam Connections	No	No	Yes	Yes	Yes	Yes
Column Base Connections	No	No	Yes	Yes	Yes	Yes
Expansion Joints	No	Yes	Yes	Yes	Yes	Yes
Construction Joints	No	No	Yes	Yes	Yes	Yes
Precast Connections	No	No	Yes	Yes	Yes	Yes

LOD 300 (Detailed Design):
- Connection types defined
- Basic detail geometry
- Load transfer mechanisms

LOD 350 (Construction Documentation):
- Detailed connection drawings
- Bolt/anchor specifications
- Weld details (if steel)
- Grouting requirements

LOD 400 (Execution):
- Shop fabrication drawings
- Installation sequence
- Torque specifications
- Quality control procedures

LOD 500 (As-Built):
- Connection inspection records
- Load test results (if required)
- Installation verification

A.3 Information Requirements by LOD

A.3.1 Geometric Information

LOD	Dimensional Accuracy	Modeling Method	Coordination Level
100	± 1000mm	Conceptual masses	None
200	± 100mm	Generic families	Basic spatial
300	± 10mm	Specific families	Full coordination
350	± 5mm	Detailed families	Interface definition
400	± 2mm	Fabrication-ready	Shop drawing level
500	As-measured	Verified geometry	As-constructed

A.3.2 Non-Geometric Information (Attributes)

Concrete Class	Generic (C25)	Specific (C30/37)	Confirmed	As-ordered	As-tested
Reinforcement	Typical (%)	Designed (bars)	Scheduled	Shop dwgs	As-installed
Dimensions	Approximate	Precise	Detailed	Fabrication	Verified
Load Capacity	Preliminary	Calculated	Confirmed	Rated	Tested
Manufacturer	—	Generic	Preferred	Specific	Actual
Cost	Conceptual	Budget	Estimated	Quoted	Actual
Weight	Approximate	Calculated	Confirmed	Fabrication	As-built
Finish	—	Specified	Detailed	Sample	As-installed
Fire Rating	—	Required	Confirmed	Certified	Inspected
Seismic Design	—	Category	Detailed	Confirmed	Verified

A.3.3 Documentation Requirements

Document Type	LOD 200	LOD 300	LOD 350	LOD 400	LOD 500
Structural Plans	Schematic	Coordinated	Permit set	Construction	Record
Sections	Key only	All major	All	Fabrication	As-built
Details	Typical	Standard	All connections	Shop	Verified
Schedules	—	Element list	Complete	Fabrication	Actual
Calculations	Preliminary	Complete	Reviewed	—	—
Specifications	Outline	Technical	Complete	—	—
Quantity Takeoffs	Rough	Detailed	Accurate	—	Actual

A.4 Quality Control and Validation

A.4.1 Model Validation Checklist by LOD

LOD 200 Validation:

- All major structural elements present
- Generic dimensions assigned
- Preliminary loads considered
- Basic coordination with architecture

LOD 300 Validation:

- Precise dimensions on all elements
- Reinforcement design complete
- Concrete classes specified
- MEP coordination complete
- Clash-free model
- Quantities extractable

LOD 350 Validation:

- All connections detailed
- Bar schedules complete
- Construction sequence defined
- Interface details coordinated
- Permit-ready documentation

LOD 400 Validation:

- Shop drawing level detail
- Fabrication information complete
- Installation sequence confirmed
- Quality control parameters defined
- Manufacturer data included

LOD 500 Validation:

- As-built dimensions verified

- Test results documented
- Deviations from design noted
- FM-ready information included
- Scan data integrated (if applicable)

A.4.2 Coordination Points with Other Disciplines

Structural Element	Architecture	MEP	PV Systems	Civil/Site
Foundations	Floor levels	Embedments	—	Drainage, utilities
Slabs	Floor finishes	Penetrations	—	Site levels
Columns	Cladding	Vertical routing	—	—
Beams	Ceiling heights	Horizontal routing	—	—
Roof	Waterproofing	HVAC units	Panel supports	Drainage
Walls	Openings, finishes	Penetrations	—	Retaining interface